STIC: Surface Temperature Initiated Closure

Overview

Surface Temperature Initiated Closure (STIC) Algorithm (Mallick et al. 2014)

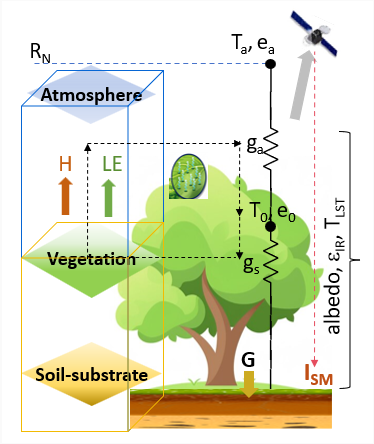
- Purpose: Single-pixel evapotranspiration (ET) estimation
- Model Type: One-source Surface Energy Balance (SEB) model
- Innovation: Uniquely integrates Land Surface Temperature (LST) directly into the Penman-Monteith formulation

Meteorological inputs

The objective of minimizing the impact of exogenous meteorological information on the performance of the product, as it only depends on

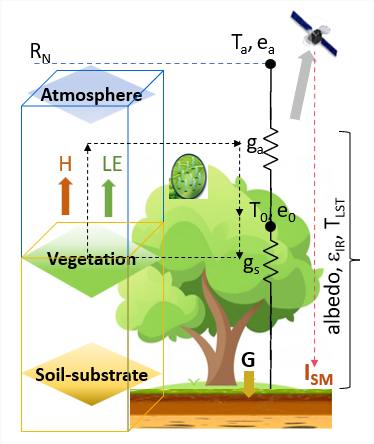
- **air temperature**
- **dewpoint temperature** (i.e. air humidity)

Analytical Framework



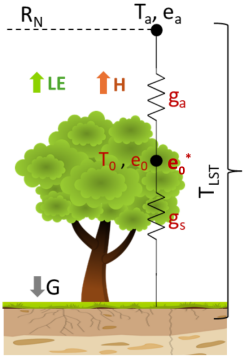
- Combines LST with aerodynamic and heat balance equations
- Determines two conductances:
  - Aerodynamic conductance (g_a) — efficiency of air transport of heat and moisture
  - Surface conductance (g_s) — vegetation canopy's ability to exchange water vapor

Analytical Framework



- Integrates:
  - LST-driven water stress index
  - Aerodynamic equations for sensible (H) and latent (LE) heat
  - Modified complementary relationship advection-aridity hypothesis
  - Shuttleworth-Wallace sparse canopy + Penman-Monteith big-leaf models

Principle



$$H = \rho c_p g_a (T_0 - T_a)$$

$$LE = \frac{\rho c_p}{\gamma} g_a (e_0 - e_a)$$

$$= \frac{\rho c_p}{\gamma} g_s (e_0^* - e_0)$$

$$= \frac{\rho c_p}{\gamma} \frac{g_a g_s}{g_a + g_s} (e_0^* - e_a)$$

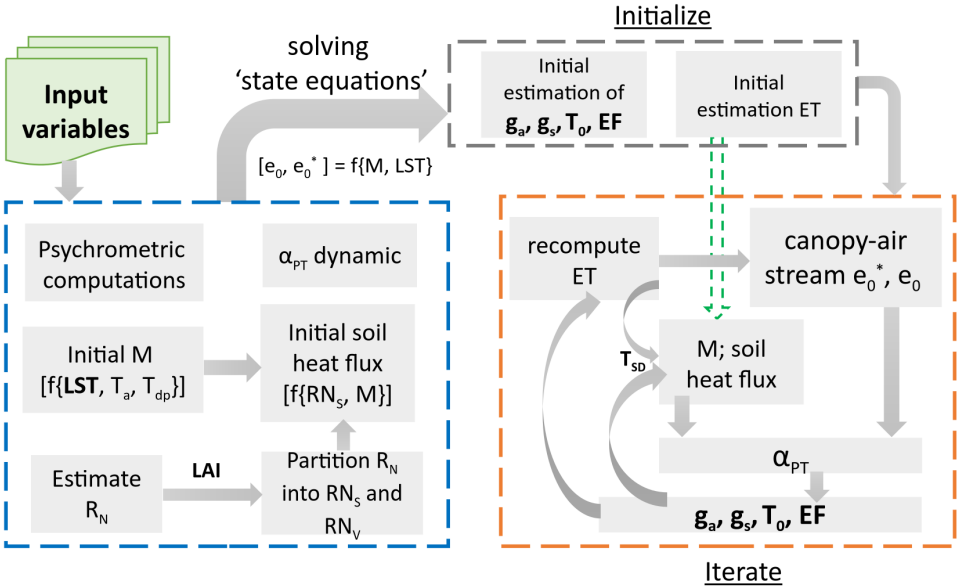
$$g_a = \frac{\gamma R_N - \rho c_p g_s (e_0^* - e_0)}{\gamma \rho c_p (T_0 - T_a)}$$

$$g_S = \frac{\gamma g_a \{R_N - H\}}{\rho c_p g_a (e_0^* - e_a) - \gamma \{R_N - H\}}$$

Dry ($T_0 >> T_a$)

Wet ($T_0 \approx T_a$)

Algorithm


Initialize
Input variables
solving 'state equations'
[e0, e0*] = f{M, LST}
Initial estimation of ga, gs, T0, EF
Initial estimation ET
Psychrometric computations
αPT dynamic
Initial M [f{LST, Ta, Tdp}]
Initial soil heat flux [f{RNs, M}]
Estimate RN
LAI
Partition RN into RNs and RNv
recompute ET
canopy-air stream e0*, e0
TSD
M; soil heat flux
αPT
ga, gs, T0, EF
Iterate

Detailed algorithm

